Sericulture Component V5

Component list with prices and use cases

slno	component name	quantity	price	use case	sericulture/poultry
1.0	Raspberry Pi 5, 4GB RAM	1	Rs. 13,409	Main edge computer for sensor acquisition, local processing, AI pipeline, data logging and control logic.	Both
2.0	Official Raspberry Pi AI HAT+ 26 TOPS – Hailo-8 AI Accelerator Board	1	Rs. 11,309	Edge AI accelerator for YOLO-based silkworm detection and compatible vision inference.	Both
3.0	Raspberry Pi 27W USB-C Power Supply – 5.1V/5A, India Plug	1	Rs. 1,204	Stable power supply for Raspberry Pi 5.	Both
4.0	SanDisk Ultra microSDXC 128GB – SDSQUAB-128G-GN6MN	1	Rs. 2,649	Raspberry Pi OS, application software and local event/log storage.	Both
5.0	Sensirion SHT40 Temperature & Humidity Sensor Module – I²C	1	Rs. 698.85	Measure shed temperature and relative humidity.	Both
6.0	Arducam OV2710 2MP USB UVC Camera – 1080p	1	Rs. 4,199.99	USB video source for silkworm movement, activity and classification analysis on Raspberry Pi 5.	Both
7.0	Waveshare 6-Channel Relay Module – Industrial, Isolated Protections	1	Rs. 4,399.99	Prototype switching interface for fan, mister/pump and spare actuator channels.	Both
8.0	Magnetic Float Level Switch – Vertical Mount, Reversible NO/NC, 2-Wire	1	Rs. 122	Water-level safety interlock to prevent dry-running of the mister/pump.	Sericulture
9.0	Dupont Jumper Wire Combo – 40Pcs each M-M, M-F, F-F, 20cm, 2.54mm	1	Rs. 150	Temporary Raspberry Pi GPIO, sensor and indicator wiring.	Both
10.0	5mm Red LED – Diffused	10	Rs. 4/each	Fault and alarm status indication.	Both
11.0	5mm Yellow LED – Diffused	10	Rs. 4/each	Warning and attention status indication.	Both
12.0	5mm Green LED – Diffused	10	Rs. 4/each	System-ready and normal-operation indication.	Both
13.0	220Ω 1/4W Resistor	30	Rs. 0.39/each	Current limiting for the 30 indicator LEDs	based on Rs. 39 per 100-piece pack.
14.0	3.3V–5V Active Buzzer Module	1	Rs. 27	Local audible alarm for critical faults and safety events.	Both
15.0	SPDT ON-OFF-ON Toggle Switch – 3A	4	Rs. 41.30	Hardware Auto/Manual mode selection for prototype control testing.	Both
16.0	Momentary Push Button Switch – 2-Pin, 5mm	10	Rs. 2/each	Reset and alarm-acknowledgement input testing.	Both
17.0	2-Pin + 3-Pin 5.08mm PCB Screw Terminal Block Set	1	Rs. 35/set	Removable, secure low-voltage connections on the final prototype board.	Both
18.0	5 × 7 cm Double-Sided Universal FR4 PCB Prototype Board – 2.54mm	3	Rs. 32	Solder the validated low-voltage wiring into the final prototype circuit board.	Both
19.0	I2S Digital Microphone Array	1	Rs. 3,289	Detect poultry acoustic events such as coughing, sneezing and abnormal breathing/CRD-related signals.	Poultry
20.0	Heater-side electrical protection	As required	Rs. —	MCB/fuse/RCBO and appropriate wiring/protection for the actual heater load.	Poultry

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21.0	Official Raspberry Pi 5 Active Cooler	1	Rs. 531	Cooling for Raspberry Pi 5 during sustained operation and AI workloads.	Both
22.0	Power Bank for Raspberry Pi 5 – 5V Output	1	Rs. 1,699	Backup power source for Raspberry Pi 5 with 5V output.	Both
23.0	Daylight White LED Strip – 12V, 5m, 6000K, Waterproof	1	Rs. 517	Night-time illumination for silkworm classification and movement/behavior analysis.	Sericulture
24.0	Micro HDMI to HDMI Cable	1	Rs. 189	Connect Raspberry Pi 5 micro-HDMI output to the 5-inch HDMI display.	Both
25.0	Solder Wire and Soldering Paste	2	Rs. 154/set	Soldering and finalizing prototype PCB and wiring connections.	Both
26.0	Resistors – 1kΩ, 2kΩ, 3kΩ, 5kΩ	10 each	Rs. 0.39/each	Prototype pull-up, pull-down, bias and voltage-divider circuits.	Both
27.0	LM2596 DC-DC Step-Down Voltage Regulator Module	2	Rs. 110/each	Provide adjustable regulated DC voltage for prototype electronics and auxiliary loads.	Both
28.0	Digital Multimeter	1	Rs. 799	Measure voltage, current, resistance and continuity during wiring and troubleshooting.	Both
29.0	USB Microphone for Raspberry Pi	1	Rs. 294	USB audio input for acoustic testing and future poultry sound-monitoring development.	Poultry
30.0	5-Inch HDMI LCD Touchscreen – 800×480	1	Rs. 2,359	Local kiosk/display for live system status and monitoring.	Both
31.0	Heat Sink for Hailo-8L M.2 Module	1	Rs. 1,711	Additional passive cooling for Hailo M.2 AI accelerator hardware.	Both
32.0	Screwdriver Set + Wire Cutting Plier	1 each	Rs. 134 + Rs. 199	Assembly, tightening, wiring and cutting during hardware integration.	Both
33.0	Utility Cutting Knife	1	Rs. 629	Cutting insulation, cable sleeves, packaging and prototype materials during assembly.	Both